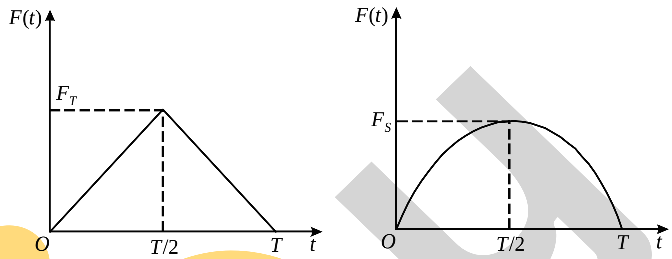
Physics Galaxy

JEE Main Level Practice Test-2
for
JEE & NEET Aspirants

Topic : NLM & FRICTION **Time: 75 Min** **Marking: +4 –1**

Section - A : MCQs with Single Option Correct

1. A car of mass m is accelerated by applying a triangular force profile shown in figure (a) same speed achieved at $t = T$ sec with a sinusoidal force profile, $F(t) = F_S \sin\left(\frac{\pi t}{T}\right)$,



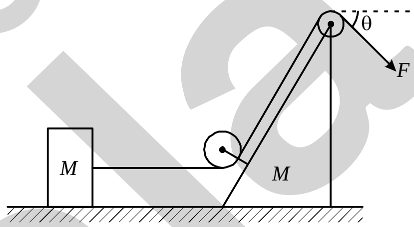
(A) Speed of the car at $t = T$ sec is $\frac{TF_T}{2m}$

(B) Speed of car at $t = T$ sec is $\frac{TF_T}{4m}$

(C) $F_S > F_T$

(D) $F_S < F_T$

2. In the arrangement shown, pulleys and string are massless and all the surfaces are frictionless. Wedge and block have equal mass M. The string is pulled by a constant force F at an angle θ with horizontal. The block is pulled horizontally by the string and that the wedge has pure translation. Then relative acceleration between the block and the wedge is :



(A) $\frac{F\cos\theta}{2M}$ (B) $\frac{F\cos\theta}{M}$ (C) $\frac{F}{M}(2-\cos\theta)$ (D) $\frac{F}{2M} - g$

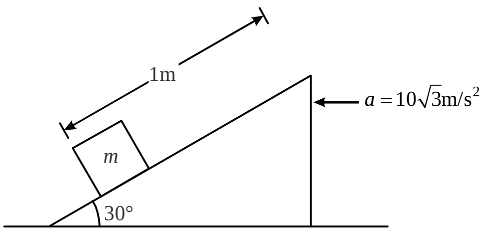
3. Two identical particles A and B, each of mass m, are interconnected by a spring of stiffness k. If the particle B experiences an external force F along the line of spring and outward (as shown) and the elongation of the spring at this instant is x, the acceleration of particle B relative to particle A is equal to :

(A) $\frac{F}{2m}$ (B) $\frac{F-kx}{m}$ (C) $\frac{F-2kx}{m}$ (D) $\frac{kx}{m}$

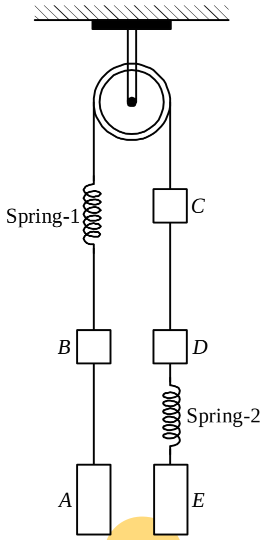
4. In the figure, the wedge is pushed with an acceleration of $10\sqrt{3}$ m/s^2. It is seen that the block starts climbing up on the smooth inclined face of the wedge. What will be the time taken by the block to reach the top? ($g = 10\text{ms}^{-2}$).

(A) $\frac{2}{\sqrt{5}}s$ (B) $\frac{1}{\sqrt{5}}s$

(C) $\sqrt{5}s$ (D) $\frac{\sqrt{5}}{2}s$



5. The system shown below is initially in equilibrium. Match the entries of Column-I with the entries of Column-II. The masses of pulleys, strings and springs are neglected $m_A = 3m$, $m_B = 3m$, $m_C = 2m$, $m_D = 2m$, $m_E = 2m$.



Column-I	Column-II
(A) Just after the spring 2 is cut, the block D	(P) Accelerates up
(B) Just after the spring 2 is cut, the block B	(Q) Accelerates down
(C) Just after the spring 2 is cut, the block A	(R) Momentarily at rest
(D) Just after the string connecting A and B is cut, the block D	(S) Moves up with acceleration g

(A) A-P, B-R, C-R, D-R (B) A-P, B-S, C-Q, D-R (C) A-R, B-Q, C-S, D-R (D) A-P, B-R, C-R, D-S

6. A small child tries to moves a large rubber toy placed on the ground by applying a force in a direction at an angle to horizontal obliquely upward. The toy does not move but gets deformed under her pushing force ($\vec{F}$). Then :

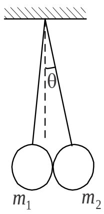
(A) The resultant of the pushing force ($\vec{F}$), weight of the toy, normal force by the ground on the toy and the frictional force is zero

(B) The normal force by the ground is equal and opposite to the weight of the toy

(C) The pushing force ($\vec{F}$) of the child is balanced by the equal and opposite frictional force

(D) The pushing force ($\vec{F}$) of the child is balanced by the total internal force in the toy generated due to deformation

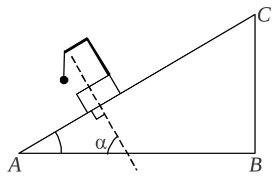
7. Two spherical objects each of radii R and masses m_1 and m_2 are suspended using two strings of equal length L as shown in the figure ($R << L$). The angle, θ which mass m_2 makes with the vertical is approximately:



(A) $\dfrac{m_1 R}{(m_1+m_2)L}$ (B) $\dfrac{2m_1 R}{(m_1+m_2)L}$

(C) $\dfrac{2m_2 R}{(m_1+m_2)L}$ (D) $\dfrac{m_2 R}{(m_1+m_2)L}$

8. A simple pendulum is attached to the block which slides without friction down an inclined plane (ABC) having an angle of inclination α as shown. While the block is sliding down the pendulum oscillates in such a way that at its mean position the direction of the string is:



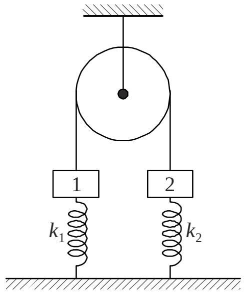
(A) at angle α to the perpendicular to the inclined plane AC

(B) parallel to the inclined plane AC

(C) vertically downwards

(D) perpendicular to the inclined plane AC

9. Two blocks (1 and 2) of equal mass m are connected by an ideal string (see figure shown) over a frictionless pulley. The blocks are attached to the ground by springs having spring constants k_1 and k_2 such that $k_1 > k_2$.Initially, both springs are unstretched. The block 1 is slowly pulled down a distance x and released. Just after the release the possible values of the magnitude of the acceleration of the blocks a_1 and a_2 can be :



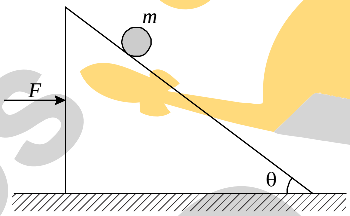
(A) Either $\left(a_1 = a_2 = \dfrac{(k_1+k_2)x}{2m}\right)$ or $\left(a_1 = \dfrac{k_1 x}{m} - g \text{ and } a_2 = \dfrac{k_2 x}{m} + g\right)$

(B) $\left(a_1 = a_2 \dfrac{(k_1+k_2)x}{2m}\right)$ only

(C) $\left(a_1 = a_2 = \dfrac{(k_1-k_2)x}{2m}\right)$ only

(D) $\left(a_1 = a_2 = \dfrac{(k_1-k_2)x}{2m}\right)$ or $\left(a_1 = a_2 = \dfrac{(k_1 k_2)x}{(k_1+k_2)m} - g\right)$

10. In the given figure, the wedge is acted upon by a constant horizontal force F. The wedge is moving on a smooth horizontal surface. Find ratio of reaction force on ball of mass 'm' by the wedge when F is acting, and when wedge is kept at rest will be: (assuming no friction between the wedge and the ball) :



(A) $\sec^2\theta$ (B) $\cos^2\theta$ (C) 1 (D) $\cot^2\theta$

11. A force F of constant magnitude acts on a block of mass 'm' and $t = 0$, initially along positive x-direction. The direction of force F starts rotating with a constant angular velocity ω in anticlockwise sense. Then the magnitude of velocity of the block as a function of time 't' is (initially block was at rest) :

(A) $\dfrac{F}{m\omega}\sin\left(\dfrac{\omega t}{2}\right)$ (B) $\dfrac{2F}{m\omega}\sin\left(\dfrac{\omega t}{2}\right)$ (C) $\dfrac{F}{2m\omega}\sin\left(\dfrac{\omega t}{2}\right)$ (D) $\dfrac{2F}{3m\omega}\sin\left(\dfrac{\omega t}{2}\right)$

12. A block of weight W rest on a horizontal floor with coefficient of static friction μ. It is desired to make the block move by applying minimum amount of force. The angle θ from the horizontal at which the force should be applied and magnitude of the force F are respectively :

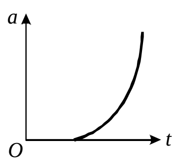
(A) $\theta = \tan^{-1}(\mu)$, $F = \dfrac{\mu W}{\sqrt{1+\mu^2}}$

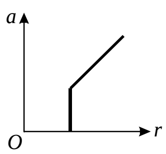
(B) $\theta = \tan^{-1}\left(\dfrac{1}{\mu}\right)$, $F = \dfrac{\mu W}{\sqrt{1+\mu^2}}$

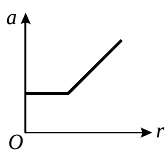
(C) $\theta = 0$, $F = \mu W$

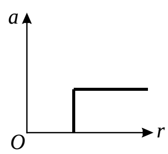
(D) $\theta = \tan^{-1}\left(\dfrac{\mu}{1+\mu}\right)$, $F = \dfrac{\mu W}{1+\mu}$

13. A block is placed on a rough horizontal plane. A time dependent horizontal force $F = kt$ acts on the block, where k is a positive constant. The acceleration-time graph of the block is :

(A)


(B)


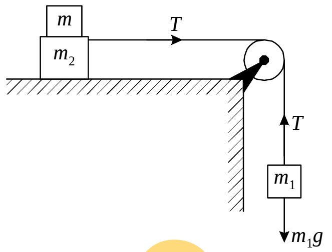
(C)


(D)


14. A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is :

(A) $\frac{1}{6}m$ (B) $\frac{2}{3}m$ (C) $\frac{1}{3}m$ (D) $\frac{1}{2}m$

15. The masses $m_1 = 5$ kg and $m_2 = 10$ kg, connected by an inextensible string over a friction less pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum weight m that should be put on top of m_2 to stop the motion is :



(A) 18.3 kg (B) 27.3 kg (C) 43.3 kg (D) 10.3 kg

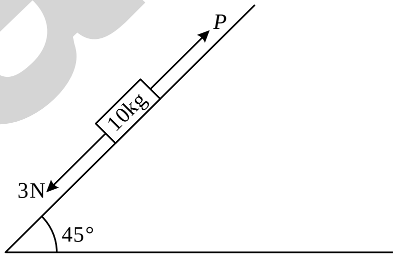
16. A given object taken n times more time to slide down a 45° rough inclined plane as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is :

(A) $\sqrt{1-\frac{1}{n^2}}$ (B) $1-\frac{1}{n^2}$ (C) $\frac{1}{2-n^2}$ (D) $\sqrt{\frac{1}{1-n^2}}$

17. A body of mass 2 kg slide down with an acceleration of 3 m/s^2 on a rough inclined plane having a slope of 30°. The external force required to take the same body up the plane with the same acceleration will be : (g = 10 m/s^2) :

(A) 4 N (B) 14 N (C) 6 N (D) 20 N

18. A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P, such that the block does not move downward ? [Take g = 10 ms^{-2}]

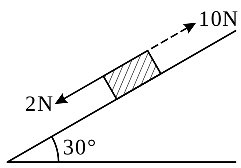


(A) 32 N (B) 25 N (C) 23 N (D) 18 N

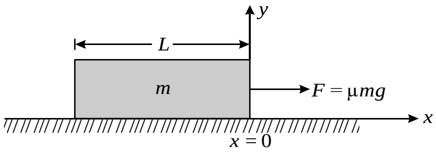
19. A block kept on a rough inclined plane, as shown in the figure, remains at rest upto a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction between the block and the plane is : [Take g = 10 m/s^2]

(A) $\frac{2}{3}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{4}$ (D) $\frac{1}{2}$



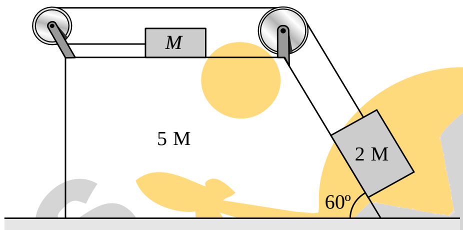
20. An uniform bar of mass m and length L lies on a horizontal floor as shown in the figure. Region $X > 0$ has coefficient of friction μ while Region $X < 0$ is smooth. A constant external horizontal force $F = \mu mg$ starts pulling the bar as shown in the figure. Speed of block when its rear end crosses origin is :



(A) $\sqrt{\mu gL}$ (B) $\sqrt{\frac{\mu gL}{2}}$ (C) zero (D) $\frac{\sqrt{\mu gL}}{2}$

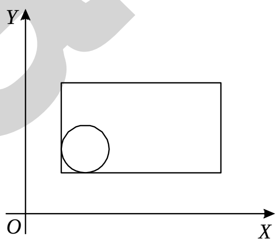
Section- B: INTEGER Answer Type Questions

21. In the system shown, the acceleration (m/s^2) of the wedge of mass $5M$ is (there is no friction anywhere) :

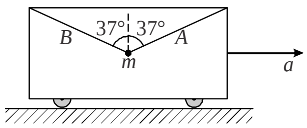


22. The maximum magnitude (in N) which is possible by adding four forces of magnitudes 1N, 3N, 9N and 10N is :

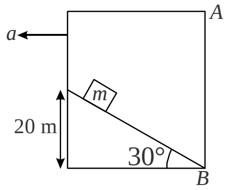
23. A solid sphere of mass 2kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V} = 5t\hat{i} + 2t\hat{j}$ m/s. Here t is the time in seconds. All surfaces are smooth. The sphere is at rest with respect to the cube. What is the magnitude of total force (in N) exerted by the sphere on the cube? [$g = 10$ m/sec^2 & $\hat{j}$ is the unit vector along vertically upward direction].



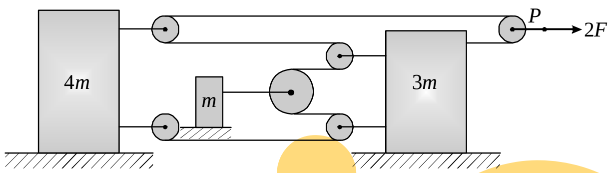
24. A ball is suspended from the accelerating car by two cords 'A' and 'B'. The acceleration 'a' of the car which will cause the tension in 'A' to be twice the tension in B, then $a = 10y$ & g = 10m/s^2. Find the value of y.



25. A block can slide along incline plane as shown in figure. The acceleration of chamber is $\frac{2}{\sqrt{3}}g$ m/s^2 horizontally as shown. The time when the block collide with the vertical wall AB of the chamber is t then find t^2. (take g = 10 m/s^2).

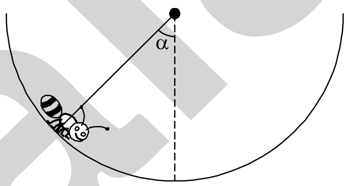


26. In the setup shown, blocks of masses m, 3m and 4m are placed on a frictionless horizontal surface and the free end 'P' of the thread is being pulled by a constant force 2F. The acceleration of the free end P is KF/m. Find K.



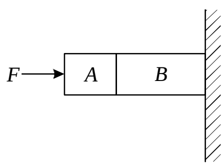
27. The minimum force required to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2. If the inclined plane makes an angle θ from the horizontal such that $\tan\theta = 2\mu$ then the ratio $\frac{F_1}{F_2}$ is.

28. An insect crawls up a hemispherical surface very slowly. The coefficient of friction between the insect and the surface is 1/3. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of cot α so that the insect does not slip is given by.



29. A body starts from rest on a long inclined plane of slope 45°. The coefficient of friction between the body and the plane varies as $\mu = 0.3x$, where x is distance travelled down the plane. The body will have maximum speed at x (in cm) = y/30. Find y.

30. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force (in N) applied by the wall on block B is.



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Physics Galaxy

JEE Main Level Practice Test-2 for JEE & NEET Aspirants

Topic : NLM & FRICTION ANSWER KEY

Section - A : MCQs with Single Option Correct

1. (D)	**2.** (C)	**3.** (C)	**4.** (B)
5. (A)	**6.** (A)	**7.** (B)	**8.** (D)
9. (B)	**10.** (A)	**11.** (B)	**12.** (A)
13. (B)	**14.** (A)	**15.** (B)	**16.** (B)
17. (D)	**18.** (A)	**19.** (B)	**20.** (A)

Section- B: INTEGER Answer Type Questions

21. [0]	**22.** [23N]	**23.** [26N]	**24.** [4]
25. [2]	**26.** [4]	**27.** [3]	**28.** [3]
29. [100]	**30.** [120 N]		